Resolution Geometry: An Adversarial Brief on General Relativity

What GR Needs, What Was Chosen, and What an Honest Realist Must Concede

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Abstract

This is a brief, not a verdict handed down in advance. Each count against general relativity's *spacetime realism* and *universality* states the charge, the direct and indirect evidence, the best defence a GR realist can mount, and a typed verdict:

- CONCEDED: the evidence forces the concession;
- CONTESTED: the evidence is real but not decisive;
- ACQUITTED: the evidence favours GR.

The castle of evidence.

- Every GR prediction descends to local data, a selected law and relational readouts; spacetime points are null.
- GR+Λ is unique only inside a declared class, with the cubic vertex computed and all orders imported.
- The constants G and Λ are measured, not derived.
- The cosmological fit needs two non-baryonic components: baryons are 4–5% of the budget against our Pantheon+ $\Omega_m = 0.333$.
- GR predicts its own breakdown (singularity theorems) and requires an R^3 counterterm at two loops, so the "at most two derivatives" declaration fails quantum mechanically.
- The continuous holonomy of curvature is not finitely identifiable, and interiors are permanent fibres.
- Supernovae alone cannot test $D = 4$ against a DGP braneworld.

The counts on which GR is acquitted.

- Its nonlinearity is measured: the U^2 self-coupling carries $-14.33''$ of Mercury's $42.98''$ per century, and $\beta = 1 \pm 2.5 \times 10^{-5}$ excludes linear superposition by $4 \times 10^4\sigma$.
- Its radiation sector is pinned: extra channels carry less than 1.6×10^{-4} of the loss, and $|c_T/c - 1| \lesssim 3 \times 10^{-15}$.
- Its structure channel is measured: $\gamma = 1$ to 2×10^{-5}.

The verdict. GR as a predictive law is acquitted wherever it has been tested. *Substantival spacetime*, *derivability* of the field equation, and *universality* are conceded. The honest exit for realists is structural realism together with an effective-field-theory reading of GR, and this brief keeps that door open.

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1 Rules of the brief

1. **The target** is two claims: (a) spacetime is a substance whose points and full geometry are physical; (b) GR is the universal, uniquely correct law of gravity. The predictive success of GR is *not* the target, and is credited wherever it holds.
2. **Evidence** is either *direct* (computed or measured here, with code) or *indirect* (established theorems or published results, cited).
3. **The defence** is stated in its strongest form, before the verdict.
4. **Measurements exclude parameter regions, not theories.** No count claims that a competitor is eliminated as a whole.

2 The counts

Count 1: Spacetime points are not physical

- **Charge.** GR needs spacetime only as a representative.
- **Evidence (direct).** In FLRW, three lapse gauges give coordinate histories that differ by up to 0.94, while the relational $a(\tau)$ agrees with the exact solution to 10^{-7}. In Schwarzschild infall, three charts give coordinate times 40.94, 34.23 and 29.78, but the same proper time 32.409886 (*GR Without Spacetime*).
- **Evidence (indirect).** The hole argument [1]; Choquet-Bruhat–Geroch uniqueness up to isometry [2]; relational observables [3].
- **Defence.** Structural realism: what is real is the metric structure up to isometry, not bare points.
- **Verdict.** CONCEDED for substantivalism; the defence retains structural realism.

Count 2: The field equation is a human selection

- **Charge.** No observation of matter fixes the gravitational law. GR is selected under declared principles.
- **Evidence (direct).**
  - Source-remainder no-go: Minkowski and Schwarzschild share $T = 0$.
  - Linear selection: conservation plus descent leave the linearized Einstein operator, while either alone leaves two free parameters.

  - Cubic ansatz: there is one vertex each at 2, 4 and 6 derivatives (EH, GB, R^3), so "at most two derivatives" is what picks EH.

- **Evidence (indirect).** Lovelock's theorem [4] assumes locality, $D = 4$, metric-only fields and second order. All-order uniqueness [5, 6] assumes at most two derivatives.

- **Defence.** The principles are natural (locality, general covariance) and well tested, so "selection" is how all of physics works.

- **Verdict.** CONCEDED as *conditional naturality*. The law is the unique fixed point of principles that are human declarations, each testable.

Count 3: The constants are fitted, not derived

- **Charge.** G and Λ are inputs.

- **Evidence (direct).**
  - Geometry alone knows only GM/c^2: Schwarzschild is scale invariant, and Kepler fixes only GM (*Cornering*, F07).
  - Pantheon+ (1580 supernovae, full covariance) gives $\Omega_\Lambda = 0.667$, while $\Lambda = 0$ is rejected by $\Delta\chi^2 = 48$ even with free curvature.

- **Defence.** Every theory has constants. Λ is permitted by Lovelock's theorem and is not ad hoc.

- **Verdict.** CONCEDED: the values are measurements. The size of Λ remains unexplained by GR.

Count 4: The cosmological fit needs two invisible components

- **Charge.** GR plus visible matter does not fit cosmology. A non-baryonic matter component and a cosmological term are added to keep the law.

- **Evidence (direct).** Our Pantheon+ fit gives $\Omega_m = 0.333$. With $\Omega_b h^2 = 0.0224$ and $h \in [0.67, 0.73]$, baryons are $\Omega_b = 0.042$–0.050. Of the matter, 85–87% is non-baryonic, and visible matter is about 4–5% of the total budget.

- **Defence.** Dark matter is evidenced independently: CMB acoustic peaks, weak lensing, cluster dynamics and the Bullet Cluster. Λ is allowed by uniqueness.

- **Verdict.** CONTESTED. This is a real accounting fact, but the independent lines of evidence prevent a conviction. GR's cosmological success is conditional on two components it does not explain.

Count 5: GR is not universal

- **Charge.** GR cannot be the final law: it predicts its own breakdown, and its quantum completion violates the declaration that selects it.

- **Evidence (indirect).**
  - Singularity theorems [7, 8]: generic, physically reasonable data develop incomplete geodesics.
  - Two-loop non-renormalizability [9]: pure gravity diverges at two loops with a Weyl^3 (i.e. R^3) counterterm. That is exactly the 6-derivative vertex our cubic ansatz finds, and which the "at most two derivatives" declaration excludes.

- **Defence.** GR is the leading term of an effective field theory, valid far below the Planck scale. EFT practice is standard and empirically sound.
- **Verdict.** CONCEDED: universality is not provable. The defence's own EFT reading concedes that R^3 and higher terms are present in principle and are suppressed, not absent.

Count 6: Much of the geometry is inferred, not observed

- **Charge.** Interiors, topology beyond the horizon, and curvature between measurements are inferred, not observed.
- **Evidence (direct).**
  - Black-hole interiors are permanent pool fibres for exterior protocols (*Undermining Spacetime*).
  - The continuous holonomy is *not* finitely identifiable: six node planes admit holonomy angles 0.656, 0.379 and 0.932 along admissible paths (*Spectral Identifiability*, R2).
  - Only the discrete (orientation) part is certifiable from finite data.
- **Defence.** Inference to the best explanation, and interpolation under smoothness, are legitimate.
- **Verdict.** CONCEDED as *declared content*. The realist may hold it, but must declare the smoothness or path budget that fixes it.

Count 7: The equivalence principle is a resolution window

- **Charge.** Einstein's box neither grounds GR nor sees structure.
- **Evidence (direct).**
  - In orbit the tidal tensor is 1284 E, resolvable by gradiometry.
  - The box reads only the energy channel: R_{0i0j} is γ-free while $R_{ijkl} \propto \gamma$.
  - The 1911 equivalence-principle prediction ($0.876''$) is half the 1915 value ($1.751''$) (*Cornering*).
- **Defence.** The equivalence principle was always meant to be local and infinitesimal.
- **Verdict.** CONCEDED in its sharpened form (an infinitesimal gauge statement); the defence agrees.

Count 8: Dimension four is not tested by supernovae

- **Charge.** A five-dimensional braneworld fits the data as well as GR.
- **Evidence (direct).** Flat DGP against ΛCDM on Pantheon+ gives $\Delta\chi^2 = -0.44$.
- **Defence.** Growth of structure, CMB and BAO combinations disfavour self-accelerating DGP, which also has a ghost.
- **Verdict.** CONTESTED. Supernovae alone do not test $D = 4$, and the defence's probes are not executed here.

Count 9 (for the defence): GR's nonlinearity is measured

- **Charge against GR.** The nonlinear structure is chosen, not measured.
- **Evidence (direct).**
  - From Mercury's orbital elements, GR predicts $42.98''$ per century. The U^2 self-coupling, i.e. the PPN β term, contributes $-14.33''$ of it.
  - Linear superposition ($\beta = 0$) would predict $57.31''$.
  - Planetary ephemerides give $\beta - 1 = (0.2 \pm 2.5) \times 10^{-5}$ [10], excluding $\beta = 0$ by $4 \times 10^4 \sigma$.
  - The cubic vertex is computed, and its off-shell consistency is checked (32/32 gauge-covariance and 8/8 Bianchi identities).
- **Verdict.** ACQUITTED: GR's self-coupling is observed in the static weak-field limit, to 2.5×10^{-5}.

Count 10 (for the defence): radiation and structure are pinned

- **Charge against GR.** The two-mode radiative content is assumed.
- **Evidence (direct).**
  - Binary pulsars: B1913+16 gives a ratio of 0.9983 ± 0.0017 (computed), and the double pulsar 0.999963 ± 0.000063, so extra radiative channels carry less than 1.6×10^{-4} of the quadrupole loss.
  - GW170817 gives $c_T/c - 1 \in [-3.1 \times 10^{-15}, 6.5 \times 10^{-16}]$.
  - Cassini gives $\gamma = 1$ to 2×10^{-5}, and Brans–Dicke $\omega > 4 \times 10^4$.
- **Verdict.** ACQUITTED within these parameter regions. Regions such as $\omega \to \infty$ and screened sectors survive and are not eliminated.

3 What an honest GR realist must concede, and what they keep

Must concede	Keeps
Spacetime points and coordinates are not physical (Count 1)	Metric structure up to isometry: structural realism
The field equation is selected under declared principles (Count 2)	Uniqueness under those principles, which are well tested
G and Λ are measured inputs (Count 3)	Their values are stable and Lovelock-allowed
Cosmological success needs two dark components (Count 4)	Independent evidence for dark matter
GR is not provably universal and is an EFT leading term (Count 5)	Unmatched accuracy in every tested regime
Interiors, beyond-horizon topology and continuous curvature are declared or inferred (Count 6)	Inference under declared smoothness is legitimate
The equivalence principle is infinitesimal (Count 7)	It holds exactly in that form
—	Nonlinear self-coupling measured (Count 9)

Must concede	Keeps
—	Radiative and structure sectors pinned (Count 10)

4 Verdict

- **GR as a predictive law:** acquitted wherever tested, including its nonlinear self-coupling.
- **Substantival spacetime:** conceded. GR does not need it.
- **Derivability of the field equation:** conceded. The equation is the unique fixed point of human declarations.
- **Universality:** conceded. Singularity theorems and two-loop divergence show GR is not final, and the correction it needs (R^3) is exactly the one the selecting declaration excludes.
- **Dark sector and $D = 4$:** contested and open.

The honest position that survives is *structural, effective realism*: GR is the best current description of relational gravitational structure at accessible scales. Its spacetime is a representative, its equation a selection, and its domain finite.

How this brief can be wrong, and the opportunities left for realism.

- Count 4 falls if a dark-matter particle is detected with the predicted abundance. Then the dark sector is physics, not a choice.
- Count 8 falls once growth, CMB and BAO data are added to the supernovae, since those probes disfavour self-accelerating DGP.
- Count 5 falls if a UV-complete theory reduces exactly to GR with *vanishing* R^3, which would make the two-derivative declaration a theorem.
- Count 2 weakens if the declared principles are derived from deeper ones, for example entropic or holographic arguments with independent tests.

5 Reproduction and limits

Direct-evidence code.

- `rg_gr_naturality_data.py`: Pantheon+, pulsars, GW170817, Cassini.
- `rg_gr_brief_data.py`: Mercury β share and the dark-sector accounting.
- `rg_cubic_universality.py`: the cubic ansatz.
- The companion batteries cited per count.

All reruns are byte-identical.

Limits.

- Published values are used for the pulsars, GW170817, Cassini, INPOP13a and $\Omega_b h^2$.
- The supernova fits are grid profile likelihoods.
- The dark-sector fraction uses an external baryon density.
- No BAO, CMB or growth data are used.
- The indirect counts rest on cited theorems.

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